

Sandeep Singamaneni

Data Engineer | Portland, OR | sandeepsingamaneni@gmail.com | (503) 839-3884 | [LinkedIn](#)

SUMMARY:

Data Engineer with 7+ years in designing, implementing, and optimizing scalable data platforms and automation solutions using AWS, Python, SQL, and modern CI/CD frameworks. Demonstrates proficiency in serverless architectures, API development, and infrastructure-as-code technologies. Adept at enhancing data accuracy and efficiency, with a strong focus on delivering innovative solutions to drive business outcomes.

TECHNICAL SKILLS:

Programming Languages: Python, R, SQL

Google Cloud Platform (GCP): GCE, GKE, GCR, App Engine, Cloud Run, IAM, Storage, Big Query, Cloud Function, Cloud Operations

AWS: EC2, ECR, S3, Beanstalk, Redshift, Athena, DynamoDB, Lambda, Glue, EMR

Containerization: Docker, Kubernetes

CI/CD: GitLab, GitHub Actions, Azure Devops

IAC: Terraform, CloudFormation

Databases/Others: RESTful & GraphQL APIs, PySpark, Grafana, Microsoft SQL Server, PostgreSQL Server, Visual Studio, Linux OS, HTML, CSS, Flask, Tableau, Wireshark

Certifications: AWS Certified Data Engineer- Associate, Google Cloud Professional Cloud Architect, Google Cloud Associate Cloud Engineer, AWS Certified Cloud Practitioner, Microsoft Certified Fabric Analytics Engineer - Associate

PROFESSIONAL EXPERIENCE:

Data Process Engineer, Providence Health Plan (Contract), Beaverton, OR, USA

Mar 2024 – Current*

- Automated data processing workflows using Python and SQL, reducing manual intervention by 30% and improving data accuracy, resulting in more efficient reporting and analysis.
- Developed and optimized ETL pipelines for healthcare data integration across SQL Server databases, decreasing data load time by 25% and enhancing overall system performance.
- Led the creation of Tableau dashboards for PCP Profiles and other business intelligence initiatives, improving decision-making efficiency for stakeholders by 20%.
- Developed an R Shiny application to forecast significant improvements in quality measures, enabling data-driven decisions for monetary benefits and performance incentives.
- Implemented Bayesian Improved Surname Geolocation (BISG) methodology in R to infer race and ethnicity using a combination of internal data and U.S. Census data.
- Visualized healthcare metrics with Grafana and Tableau, improving stakeholder decision-making.

AWS Data Engineer, Union Bank of India, Chennai, India

Nov 2017 – Nov 2021

- Designed and implemented a high-throughput data pipeline processing 10TB of data daily using Amazon EMR, reducing processing time by 30%.
- Streamlined real-time data processing in a banking environment with Spark Streaming and Kafka, enhancing data flows and analytics for improved decision-making.
- Utilized Hadoop on AWS EMR for scalable data processing, automating EMR cluster setup with Python and Boto3, and integrated data into S3, Elasticsearch, and Hive external tables using AWS Lambda, Glue, and PySpark.
- Designed GraphQL and RESTful APIs for seamless integration across financial systems and Deployed Docker containers for scalable microservices and ensured CI/CD implementation via GitHub.
- Developed and managed 50+ data models in dbt to optimize data warehouse performance, reducing query response times by 30%, and orchestrated end-to-end data pipelines with Apache Airflow, processing 5TB+ daily transactional data for real-time banking insights.
- Developed PySpark code to transform raw data into structured format jobs on AWS EMR, optimizing data processing workflows.

Software Engineer, Wipro Ltd, Hyderabad, India

Feb 2015 – Jan 2017

- Designed and streamlined AWS (S3, EMR) data solutions resulting in a 30% growth in customer satisfaction.
- Built ETL /ELT data pipelines resulting in 25% faster data processing times.
- Streamlined data access by creating a data lake using AWS Athena, resulting in 30% faster report generation, resulting in a 40% improvement in operational efficiency and a 10% increase in customer satisfaction ratings.
- Orchestrated collaborative efforts with cross-functional teams of architects and technical consultants, resulting in successful implementation of innovative solutions, driving a 20% surge in customer satisfaction scores.

EDUCATION:

Master of Science in Computer Science | Portland State University, Portland, USA | GPA:3.91

Mar 2022 – Aug 2023

Bachelor of Technology in Computer Science | Sreenidhi Institute of Science and Technology, Hyderabad, India

PROJECTS:

Serverless Guestbook Transformation: API Gateway, Lambda, S3

- Restructured monolithic guestbook application into a serverless architecture, deploying a robust REST API with AWS Lambda and API Gateway. The front end, hosted on S3, provides a static, responsive interface. Demonstrated proficiency in serverless computing and AWS services, optimizing application performance.

Serverless Image Retrieval Web App: API Gateway, Lambda, Web Console

- Created a Lambda function and API to fetch and display random images using AWS services. Streamlined development using the AWS web console for code construction and deployment.

CAMPUS EMPLOYMENT:

IT Helpdesk Technician, Portland State University (Part-time)

Jun 2022 – Aug 2023

- Pioneered base-level IT support to 30k+ non-technical personnel within the organization.
- Installed software, modified, and repaired hardware, and resolved technical issues; increased efficiency by 15%.
- Collaborated across teams using Jira and Confluence, conducted root cause analysis, implemented solutions, and reduced response time by 25% while enhancing customer interactions without escalation.
- Increased ticket closure rate to 80% and customer satisfaction ratings to 4.9 out of 5 in one year